Modeling stimulus detection in the MT using model weights generated by neural activity

Abstract

In this study, we attempt to correlate some aspects of psychophysics with how stimuli and decision neurons interact in an artificial neural network, in particular a CNN trained on weights generated from neurons in MT. The CNN classify coherence of moving dots on a test set, and other computationally generated stimuli are then provided to as input to be classified.

Keywords:

1 Introduction

Sensory stimuli are processed through hierarchical networks. Each layer of processing contributes to the shaping of spike trains in higher cortical areas. Similarly, internal dynamics such as memory retrieval or prediction can generate spike trains in the absence of external input. Large-scale electrophysiology is a classic method of collecting data on neural activity. While other methods are able to collect a greater breadth of data, electrophysiological records are significantly less technologically cost, more accessible, and provide a much finer granularity, typically on a millisecond scale. One of the key measurable outputs of neual activity from these recordings is the generation of spike trains, which reflect neuronal firing on a more continuous scale, as shown in a sample spike train in Figure 1A.

Since cortical spiking activity is often highly variable, it has been suggested that only rate-based encoding matters in terms of neural processing [1]. However, spike patterns like burstiness are consistently similar across functionally similar cortical regions, and consistently different across different functions (e.g., sensory vs. motor vs. executive), suggesting that firing patterns do matter. In particular, we hypothesize that similarity in spike trains reflect the function of a particular type of neuron, those which are stimulus-activated and those which are decision-activated. One classic experiment with a distinct stimulus and decision component is the “moving dots” experiment, in a monkey is shown a screen with moving dots for 2000 ms, under various of trial conditions. There are signal dots and noise dots; the signal dots move in a single direction, while the noise dots point randomly. In each trial, in addition to having a direction the signal dots have coherence, e.g. percentage of dots pointing in that direction. [1]

In this study, using data collected from electrode recordings of primate neural activity in the MT from the classic “moving dots” experiment from various labs (Shadlen, Britten, Newsome)[1,3], we use a method of PCA introduced by

Kobak et al known as demixed PCA, or dPCA [2], a method of finding structures in population data using dimensionality reduction. Given peristimulus time histogram (PSTH) input, dPCA embeds firing rates into a state space on a manifold that is not condition independent, instead using a decoding axis that decomposes PSTHs into marginalized averages, and offers control over which condition we wish to decode over. In particular, the Machens study, using spike trains from Romo's "moving dot" experiment as an example, "demixes" the stimulus and decision component of PSTHs, from which encoding weights can be extracted. Here, we process datasets from Britten [3] et. al's "moving dots" experiment in the MT, extracting spike trains and their corresponding PSTHs over trials, before applying dPCA to see if our extracted weights can successfully reconstruct the original waveform of the PSTH, and find that they can do so with consistency.

Next, we design a CNN representing neurons in the MT, take the stimulus weight components in particular to tune how active each node representative of a neuron in response to classifying stimulus tasks for video classification of computationally generated moving dots, to see at which level coherence in stimuli correlates with a particular decision, and design a few more conditions (color, scale, etc) to see how well these weights can adapt to other classification tasks. Initially, we predict coherence from video samples as a form of validation, by methodically checking that the stimulus weights learned from the neurons perform well, and then use that network as a classifier for further computational experiments, providing it with other visual stimuli to examine how the network based on the MT work in terms of classification.

2 Methods

2.1 Data Preprocessing

The Britten/Newsome dataset [1] contains recordings from 52 neurons from a particular monkey, stored as a sequence of times paired to the metadata for that trial. Recordings took place over 2000 ms, where 100 ms in, the animal would be shown a "moving dot" stimulus, and have to make a decision about which direction the dots were moving in. The associated metadata tags each trial with a direction, coherence, and choice. Sample spike rasters for all trials of two sample neurons are shown in Fig. 1, where each dot represents a spike, truncated at 1000 ms for clarity, displayed on a spectrum from 25% coherence in one direction to 25% in the other. In total, there are 9, 4 in either direction and then 0 for no coherence. The figure displays a generally noisy neuron, for all levels of stimulus coherence (and likely direction as well due to the random bursting nature of the spike trains), as well as a direction selective neuron, which shows significantly higher activity for its preferred direction than anti-preferred, decreasing in a spectrum.

Firing rates are plotted as histograms of spike counts, are smoothed using convolution by a Gaussian kernel, using a sliding 50 ms bin. Sample PSTHs shown in Fig. 2 for a noisy and direction selective neuron. These PSTHs were used as input for dPCA, which was performed using a package from Kobak et. al. [2] Waveform reconstruction is based on a linear combination of dPCA components for the first ten stimuli-only components (and of course the time component).

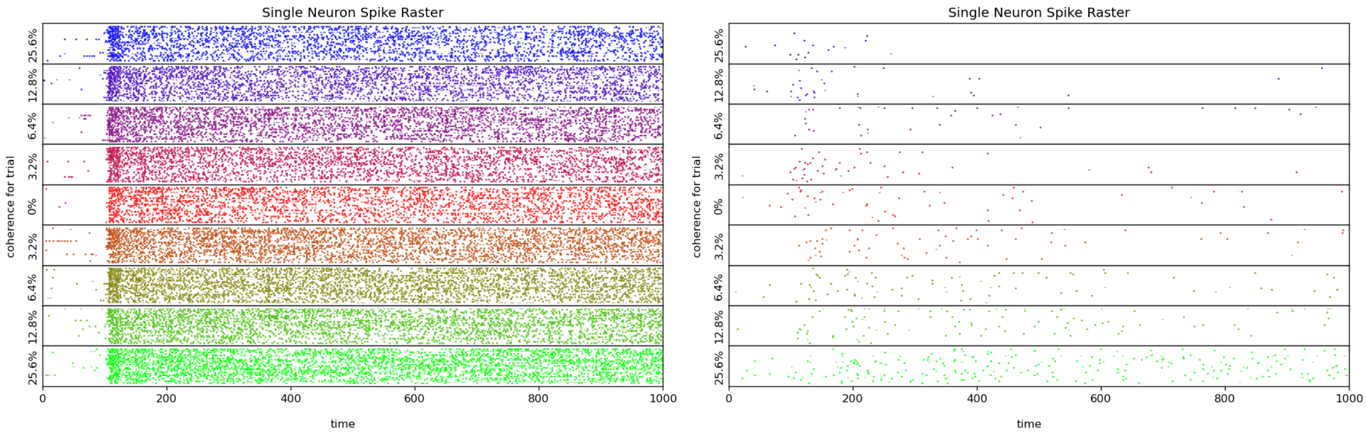

Figure 1. Spike trains plotted by coherence for all trials for two sample neurons in the Britten/Newsome dataset, (A) noisy and (B) direction selective.

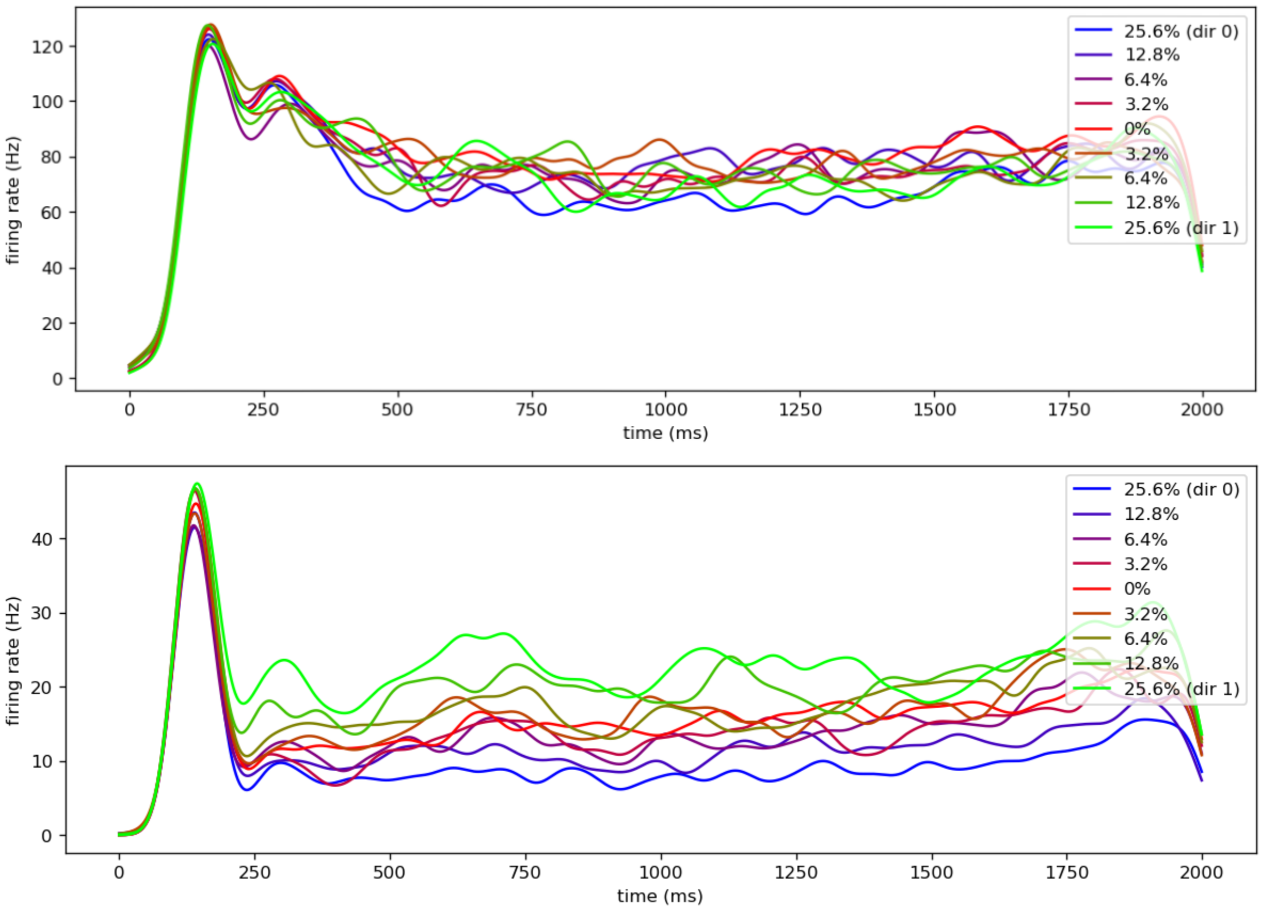

Figure 2. Sample PSTHs of coherence in two neurons in the Britten/Newsome dataset [1].

2.2 CNN Design

A CNN of 52 neurons was chosen due its capacity to process video input. Unfortunately, the entire dataset of 187 neurons could not be properly fitted in the time frame due to differences in stimulus conditions between the three animals the Britten study was performed on, although this could potentially introduce more noise than necessary into the network weights, since the original experiments were performed under different conditions, and the original paper analyzed each set of recordings separately as well. Each of the 52 stimulus weights as a scaling factor on 52 convolutional filters for each node in the CNN. The CNN architecture includes two fully connected convolutional layers designed to capture spatial and temporal features from the input video. The model is trained using Adam optimizer with a learning rate of 0.001. Training is conducted for 5-8 epochs on a dataset consisting of 600 samples, with a training-validation split of 80%-20%, and a completely separate test set is also computationally generated and used.

Video input of moving dots was generated by a script from Kiani et al. [3], rewritten in Python.Each video is made up of sequence of 88 frames representing random dot motion, with a 32 pixel by 32 pixel resolution, capturing both spatial and temporal information. Each video was in black and white. Coherence values were increased in the generated input relative to the original dataset due to the small size of the network, using a set of coherences ranging from [0.0, 0.2, 0.4, 0.6, 0.8, 1.0].

3 Results

3.1 dPCA Waveform Reconstruction

All PSTHs were used as input for dPCA, which shows several interesting patterns. The in the first three principal components for condition independent waveforms, we see that most neurons immediately spike upon neural processing of the stimulus passing to the MT and then activity decreases, indicating that the decision is made almost instantaneously; this is made more apparent in the second PC, where there is a clear cessation of neural activity before the 250 ms mark, indicating that the decision is made within 150 ms, although our sample direction selective neuron continues firing for its preferred direction for a longer duration; we will examine this in Fig. 3. The third PC, meanwhile, shows firing rates that are suppressed before increasing their spike activity; this may represent the population of anti-direction selective neurons, which inhibit neural activity. After the initial wave of inhibition wears off, their activity continues, possibly contributing to suppressing a decision.

Fig. 4 and 5 perform dPCA on waveforms corresponding to only, as well as a stimulus-decision. The first stimulus-only PC shows very obvious direction selectivity, as well as anti-selectivity. The second PC for the stimulus-decision waveform shows interesting behavior when examined closely in figure 5; when the waveforms are separated by decision, they resembled inversions of one another, suggesting that there is a significant population of neurons that are selective in either direction, and respond most highly to changes in coherence.

Fig. 6, in contrast, shows the first three principal component waveforms for dPCA factoring in only the decision. Note that the first two PCs show that the typical waveform for a decision neuron is relatively instantaneous and have little

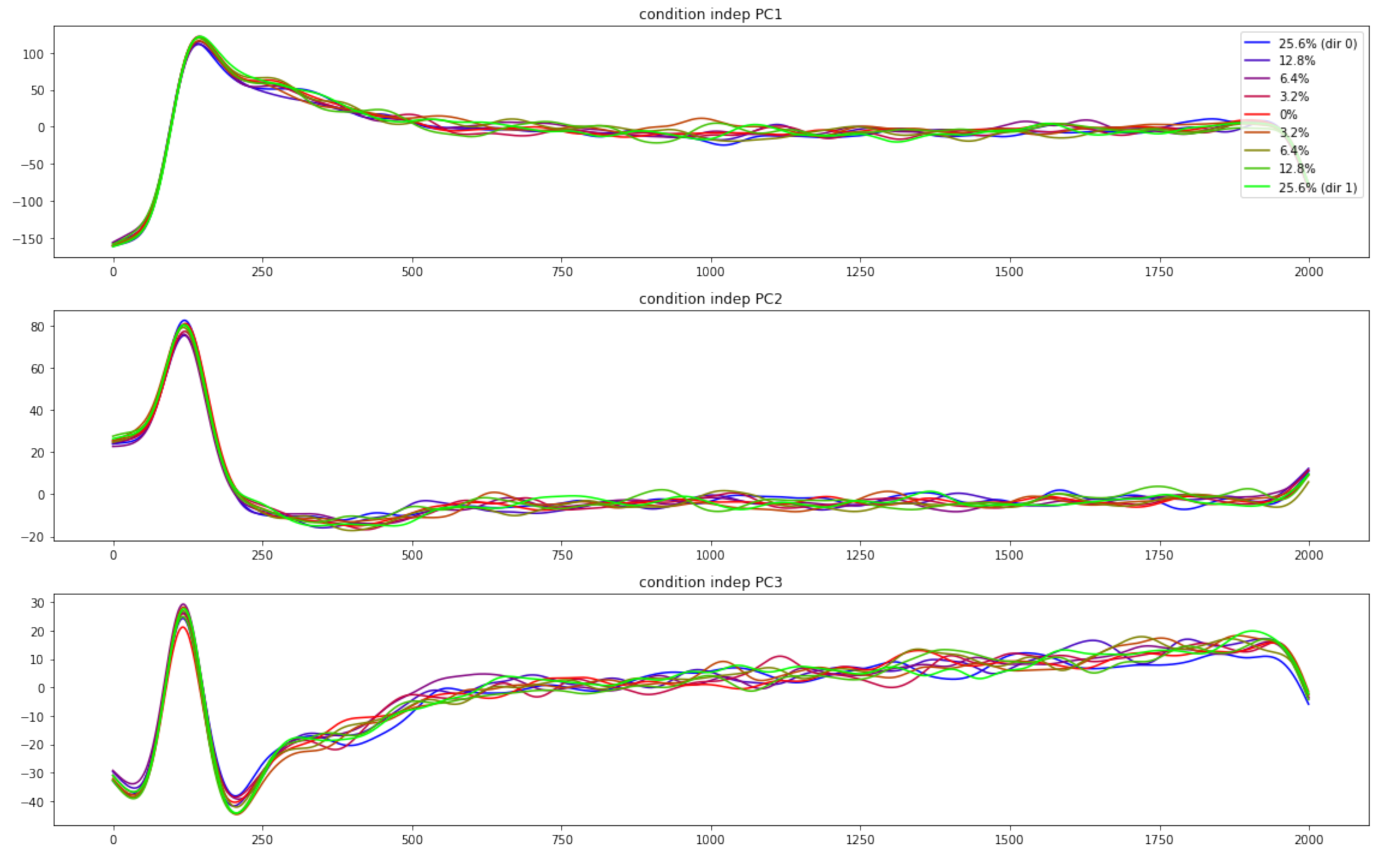

Figure 3. Waveforms of the first three PCs of condition-independent dPCA performed on PSTHs the entire dataset.

regard for direction selectivity, and it is only when we get to the third PC that we begin to see more clear interaction with direction selectivity.

Paying particular attention to the stimulus only components, we can reconstruct original waveforms using on a linear combination of dPCA components for only the first ten stimuli-only components to a high degree of success, (see Fig. 6 for examples). In general, reconstructed waveforms were slightly smoother than the original, but shared the same shape and patterns of direction selectivity (or lack thereof), firing rate, suppression/inhibition, etc. Normalizing those 10 weights for every neuron provides 52 values; using the CNN described in the Methods section, we next attempt coherence classification, using the same weights from the waveform reconstructions to determine how active each neuron is in the network.

3.2 Coherence Classification

Due to the small size of the network, number of epochs had to be set fairly low to avoid overfitting. However, with only a few epochs, training, testing, and validation sets achieved very high accuracy classification accuracy for the set of higher coherences we chose relative to the original experiment. Repeating the training with the original set of coherences [0,1.6,3.2,6.4,12.8] yields slower learning but still overall good results. The negative coherences were not implemented as the weights were based on the stimulus only-components, rather than the mixed components. The two fully connected

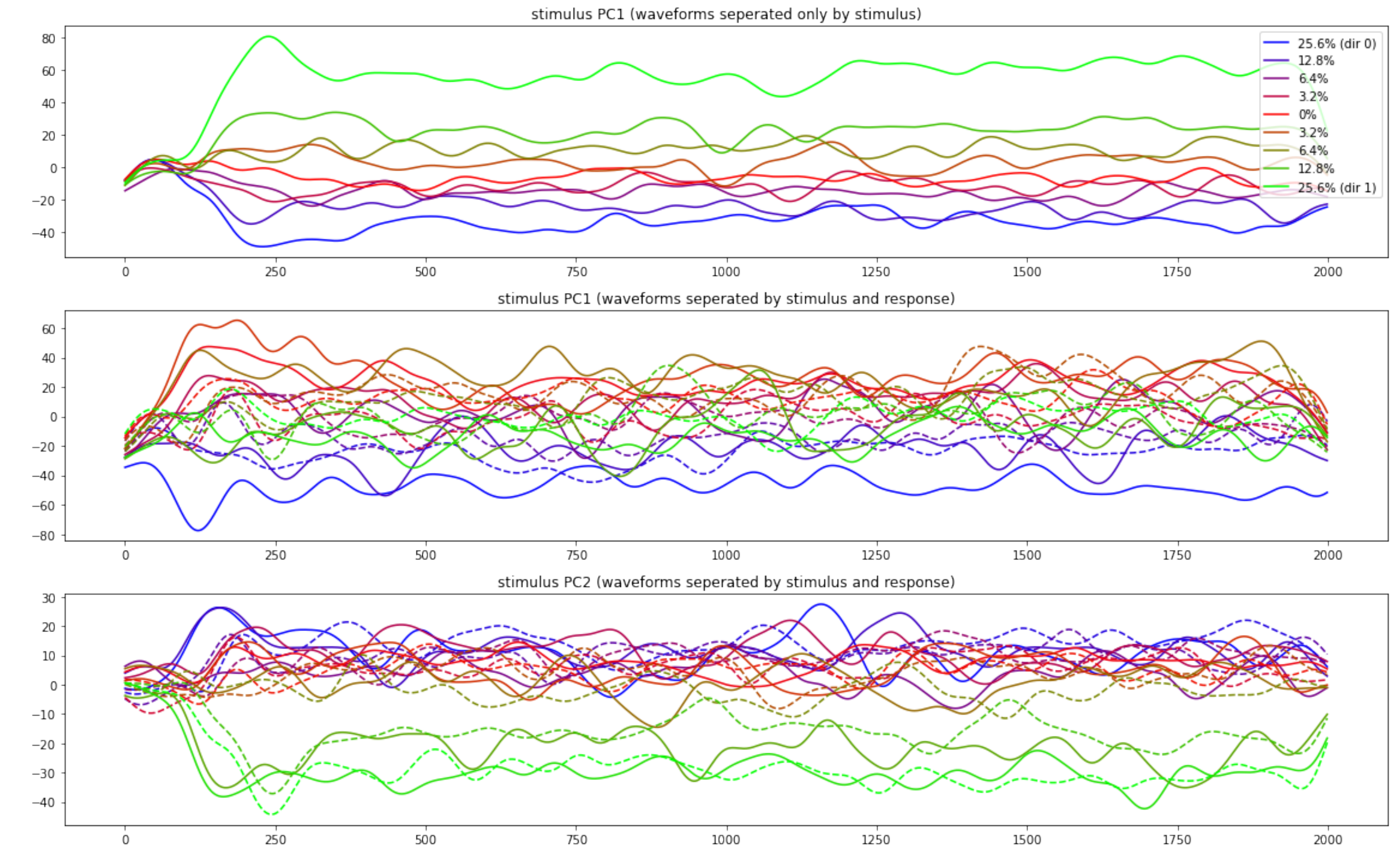

Figure 5. Waveforms of a) the first stimulus-only PC, b) the first mixed stimulus-decision PC, and c) the second stimulus-decision PC using dPCA performed on PSTHs the entire dataset. Solid lines represent decision 1, dashed lines represent 2, where 1 and 2 are decisions made on dots moving in opposing directions.

layers were meant to represent recurrent connections, which are physiologically known to exist, although here we only have figures for the first one.

The log-scaled absolute values of the heat map (Fig. 9) show extremely dense connection, which make sense due to the small size of the network. In mechanistic modeling, this also tends to be the case with connectivity matrices; real cortical circuits have far too many neurons to be captured efficiently by a model, so connectivity is often artificially dense. A non-absolute heat map (Fig. 10) shows that positive (excitatory) and negative (inhibitory) weights have a similar frequency, which agrees with the hypothesis that many neurons in the network show anti-selectivity via inhibitory connections.

4 Discussion

Our CNN seems to agree with the idea that in the MT, neurons tend to be generally directionally selective, with decisions made fairly instantaneously, and with inhibition, or anti-selectivity, playing as great a role as excitatory selectivity. Further analysis of the two fully connected layers could show a hierarchical nature, or recurrent connections, in the architecture of the MT.

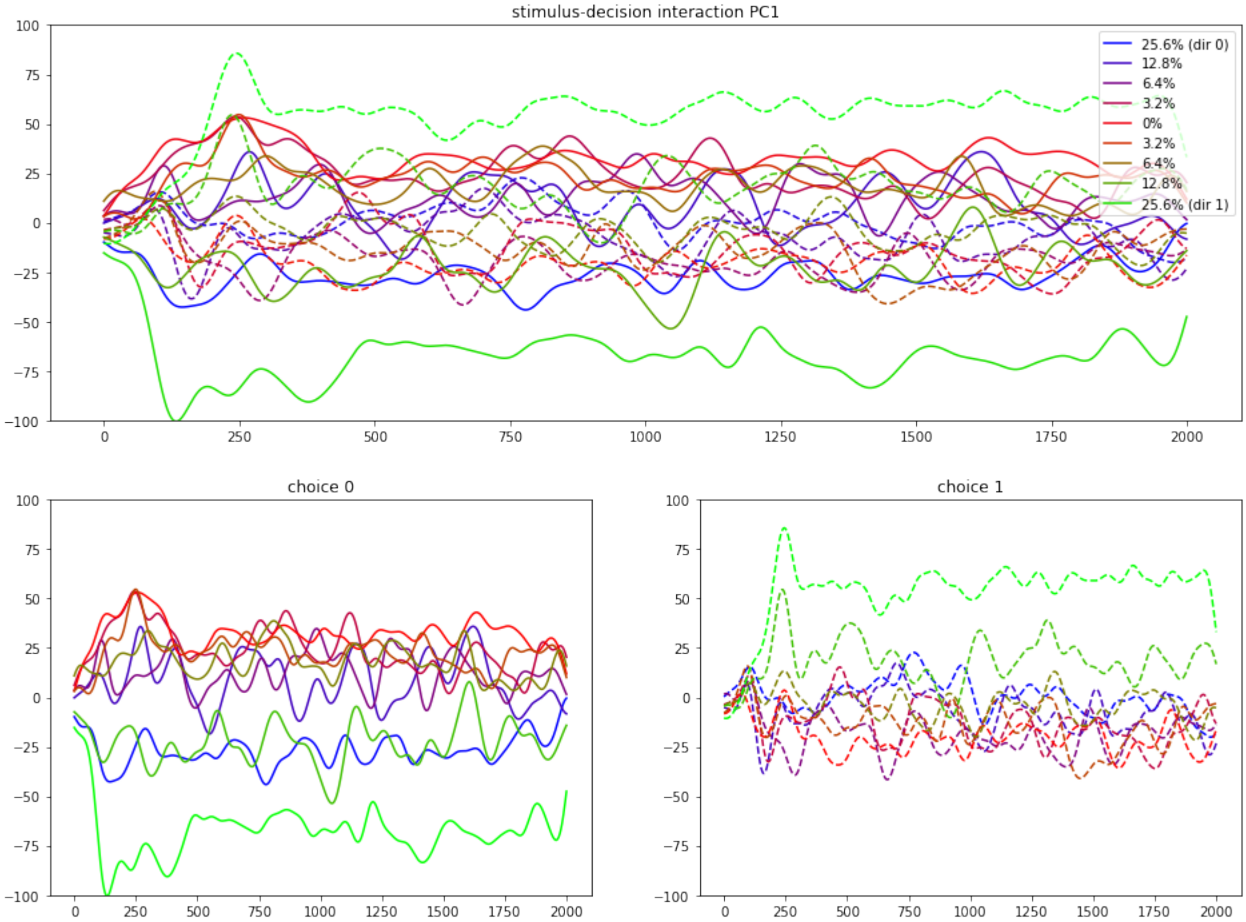

Figure 5. Closer look at the first stimulus-decision PC, which shows similar strong preference for highest coherence, and a spectrum of direction selectivity in opposing choices.

Ultimately, the small network size was the main detriment to the success of the network, which was prone to overfitting. Further work on this project could include integrating more neurons into the network, by using a normalization method to combine neurons from multiple datasets, e.g. by normalizing coherence on a scale of [1,-1] and using a L1 distance metric to choose a few coherence points between experiments performed by different labs and integrate those into the dPCA weights. In addition, classification could be improved by tagging a "decision" and "stimuli" contribution to each neuron, and potentially determining which neurons are more active in behavioral decision prediction. More sets of stimuli could be computationally generated for random dot motion, e.g. color, density, velocity, to analyze activity and adaptivity of more stimulus- or decision-leaning neurons. Finally, an interesting optimization experiment could include attempting to design a more sparse network, potentially leading to more neurobiologically accurate of cortical circuits in general.

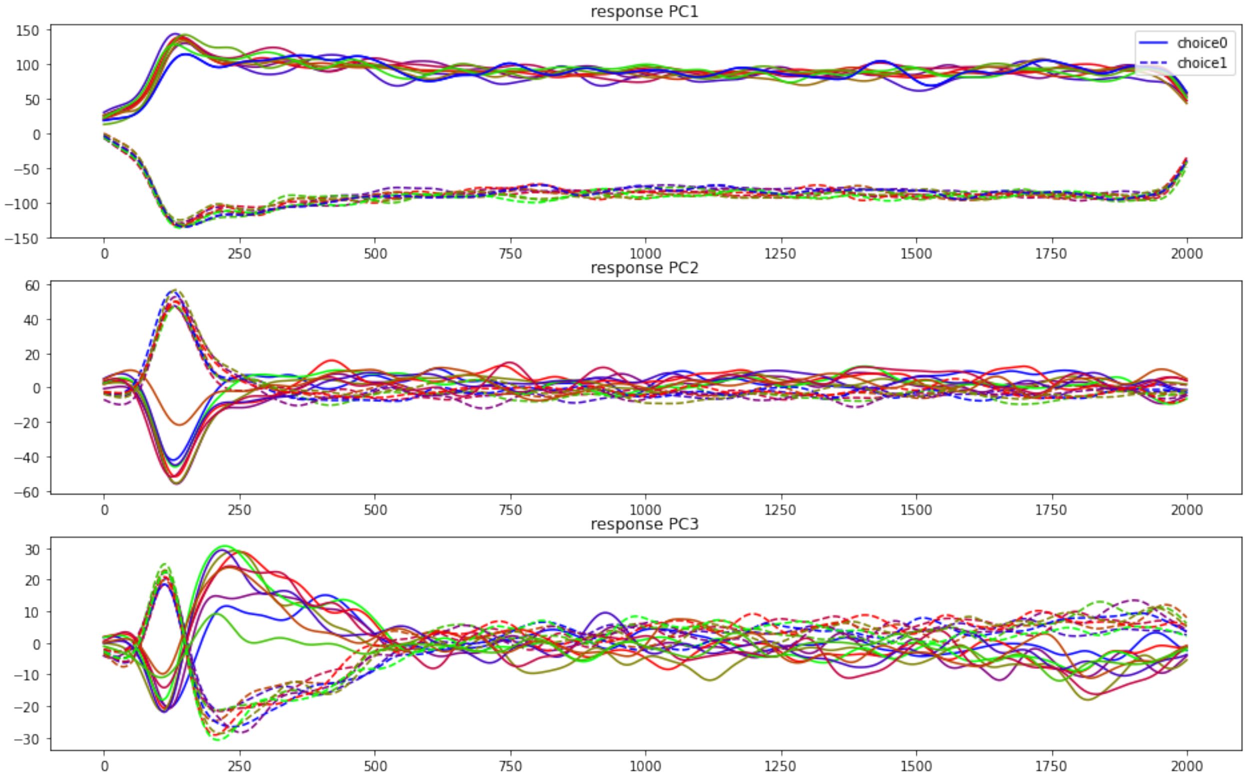

Figure 6. First three waveform PCs for decision-only dPCA. Note that the first two PCs show that the typical waveform for a decision neuron is relatively instantaneous and have little regard for direction selectivity.

5 References

[1]Britten KH, Shadlen MN, Newsome WT, Movshon JA. Responses of neurons in macaque MT to stochastic motion signals. Vis Neurosci. 1993 Nov-Dec;10(6):1157-69. doi: 10.1017/s0952523800010269. PMID: 8257671.

[2]Dmitry Kobak, Wieland Brendel, Christos Constantinidis, Claudia E Feierstein, Adam Kepecs, Zachary F Mainen, Xue-Lian Qi, Ranulfo Romo, Naoshige Uchida, Christian K Machens (2016) Demixed principal component analysis of neural population data eLife 5:e10989

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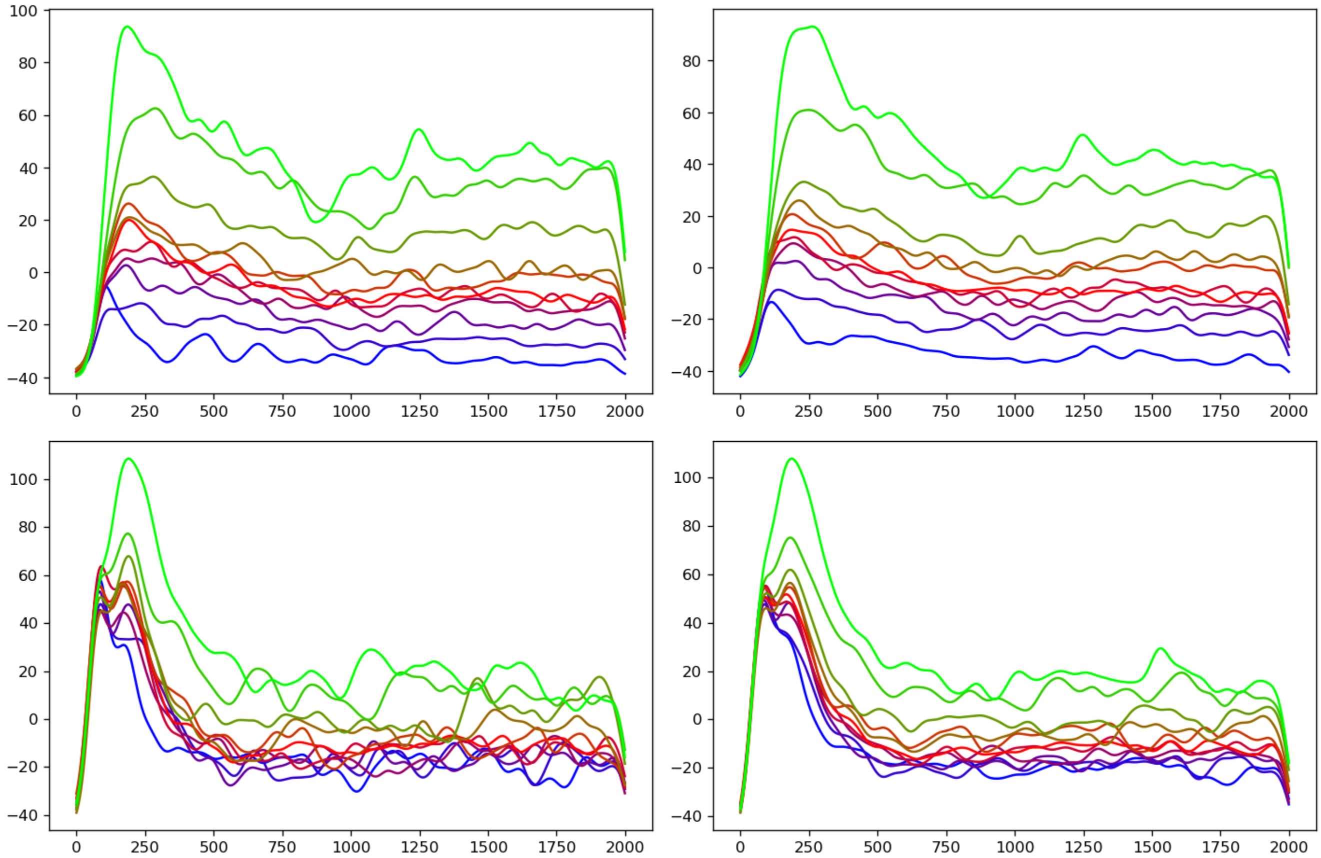

Figure 7. Sample waveform reconstructions using the first 10 PCs of the stimulus-waveform.

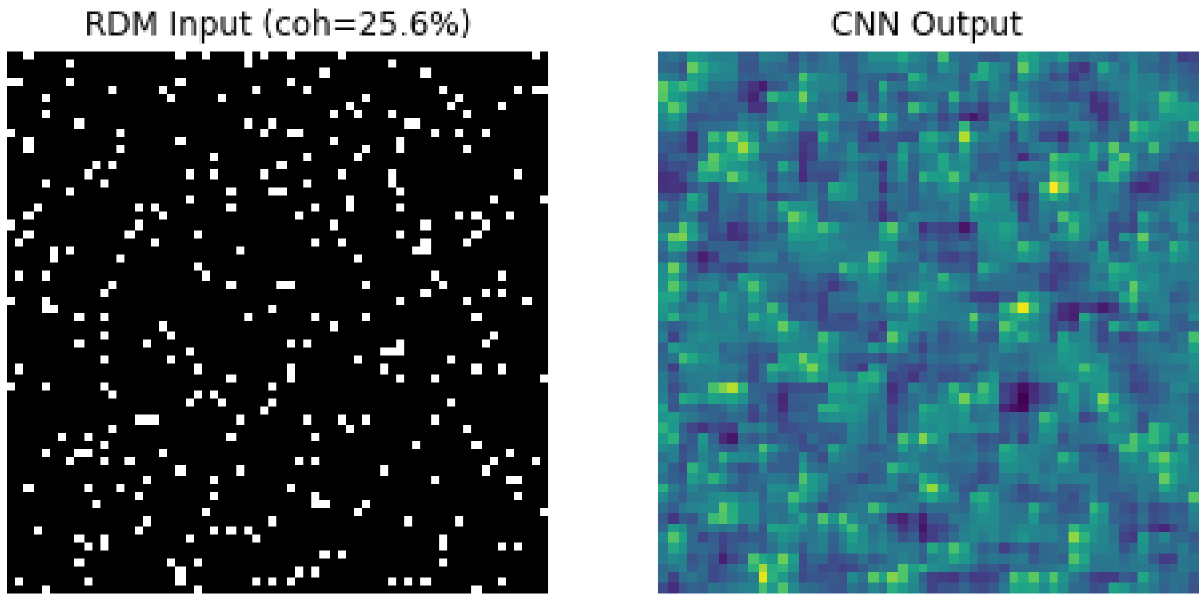

Figure 8. Sample still frame input of random dot motion to the CNN, as well as the masking layer that processes the input.

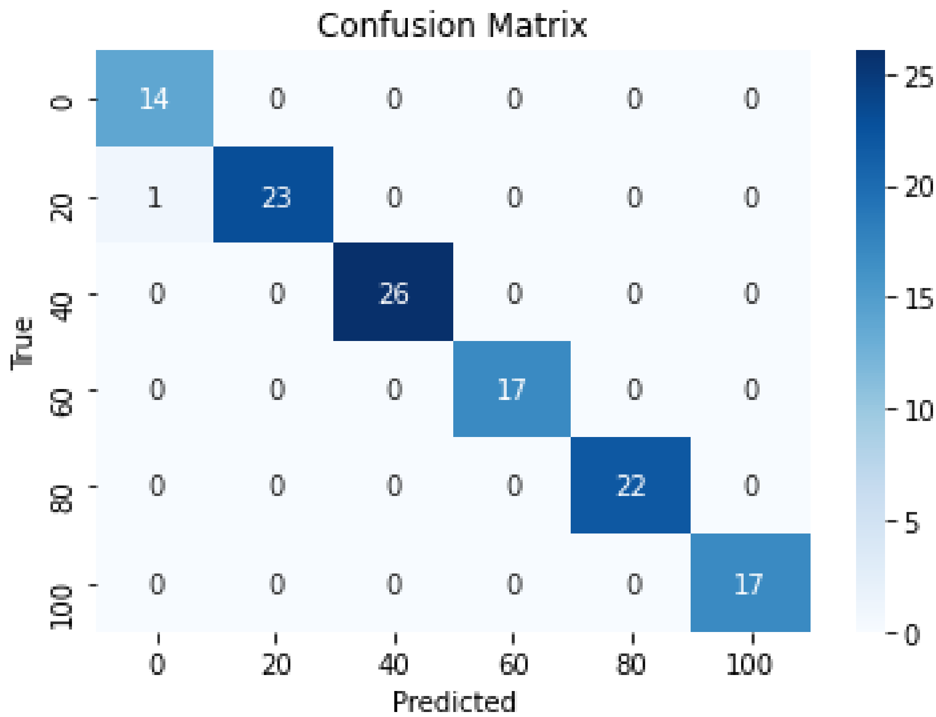

Figure 9. Confusion matrix for test set, reflecting 99.17% accuracy.

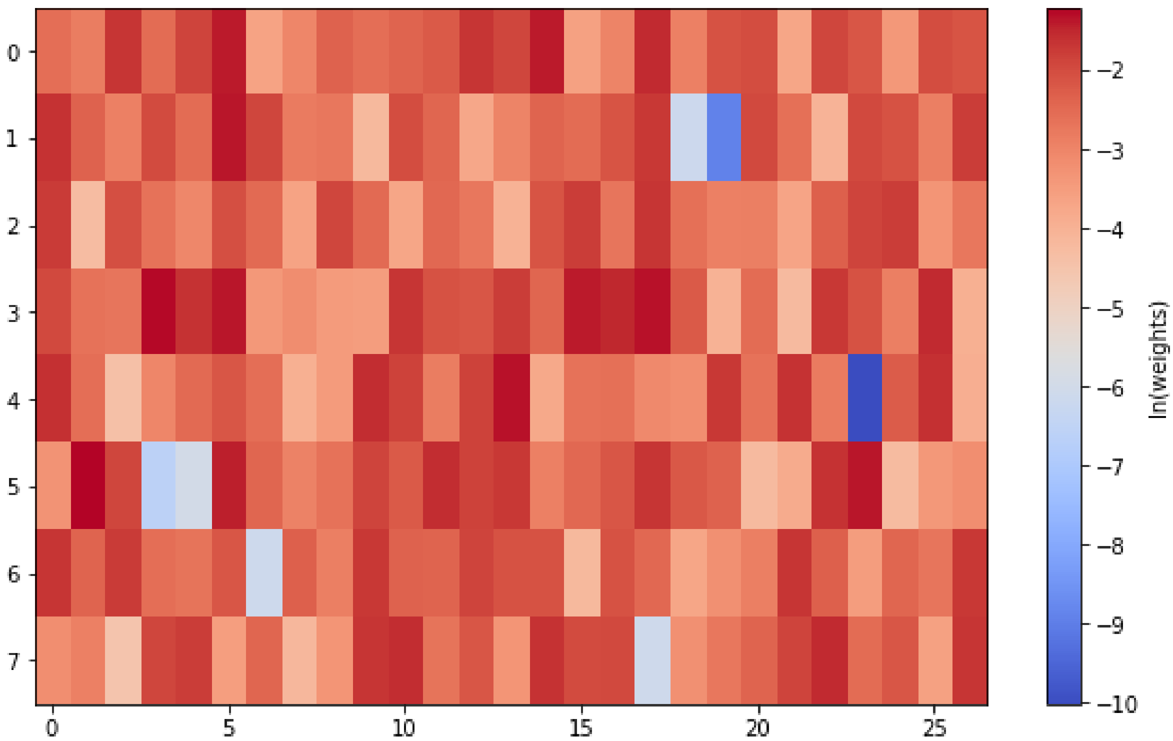

Figure 10. Log-scaled absolute weights for connectivity in the first fully connected layer.

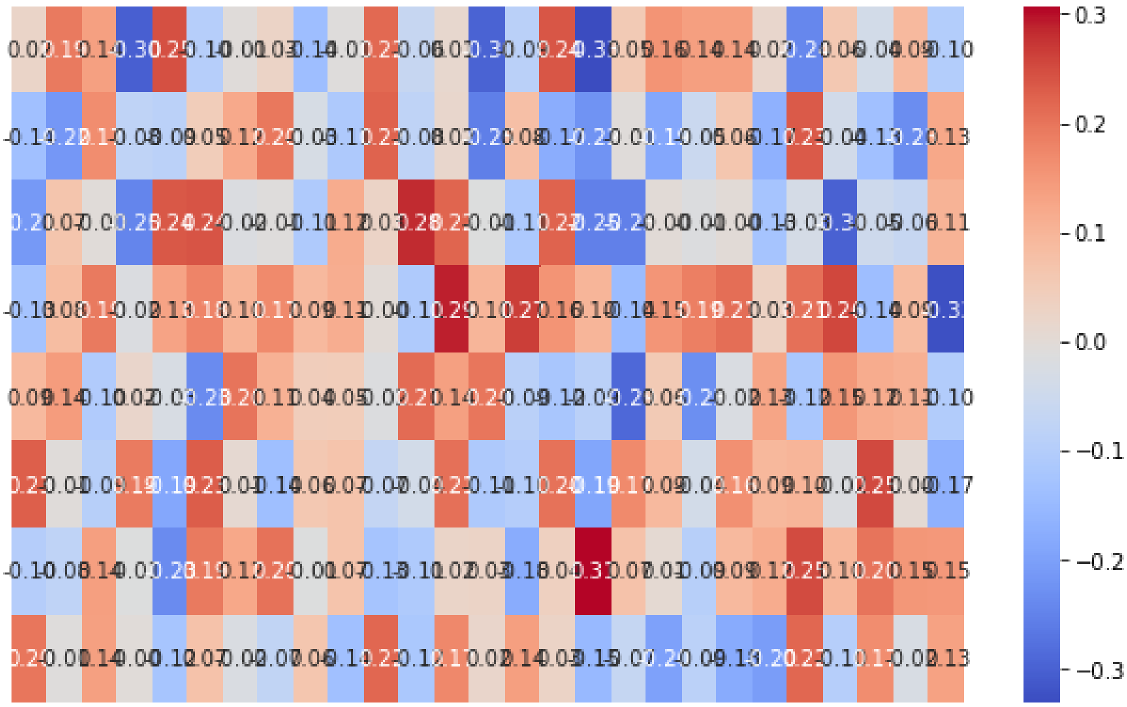

Figure 11. Heat map of weights for connectivity in the first fully connected layer.